# Understanding Unix

**NUnix · L3 MIAGE · CM1 · 90 minutes**

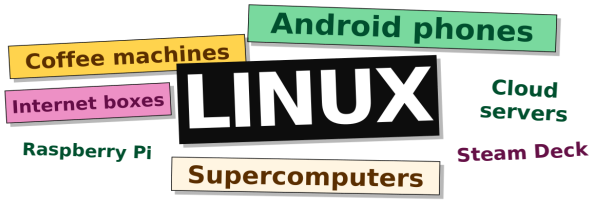
ISTIC · University of Rennes 2026–2027

Université de Rennes
Louis Ledoux



## Linux is closer than you think

Université de Rennes
Louis Ledoux




You can use it all day without opening a terminal.



## Yes, even the coffee machine

Université de Rennes
Louis Ledoux

**Android phones**
The **Linux kernel** manages processes, memory and hardware.
Google · platform architecture

**Freebox Server Pop**
A home internet box. Free publishes its **Linux source code**.
Free · firmware 4.9.10

**MARO Model One**
A coffee machine with **embedded Linux**, remote support and updates.
Toradex · May 2026

Linux can run behind a touchscreen or a web interface.



## Unix and Linux usage

Université de Rennes
Louis Ledoux

**Mobile web use**
**69.17%**
of observed mobile page views come from **Android**, built on Linux.
Statcounter · Sept 2026

**Websites**
**92.1%**
use the **Unix family**, including Linux, among sites whose OS W3Techs identifies.
W3Techs · 1 Oct 2026

**Supercomputers**
**500/500**
systems in the **June 2026 TOP500 list** use the Linux family.
TOP500 · June 2026



## Course structure

Université de Rennes
Louis Ledoux

**Three lectures of 90 minutes.**

| Lecture | Topics | Application |
|---|---|---|
| **CM1 · Unix** | Files, paths and permissions | Find files and check access |
| **CM2 · Commands** | Processes, search, pipes and redirections | Count errors in a log |
| **CM3 · Bash** | Variables, conditions, loops and functions | Write scripts with arguments and loops |

The exercises practise commands first, then scripts.



## One line, a useful result

Université de Rennes
Louis Ledoux

Which errors occur most often in **1,000 log entries**?

```
sort < errors.log | uniq -c | sort -nr | tee output.txt |
  a2ps -1 -o - | ps2pdf - rapport.pdf
```

Save the counts in `output.txt` and the report in `rapport.pdf`.

**errors.log** · first three lines

```
Database timeout
Login rejected
Connection refused
```

**output.txt** · first three counts

```
    320 Database timeout
    210 Login rejected
    140 Connection refused
```

`a2ps`: text → PostScript
`ps2pdf`: PostScript → PDF

By the end of this course, you will understand this command.

## Why use a script?

Université de Rennes
Louis Ledoux

A **script** stores shell commands in a text file.

Save the report command in a file.
Run that file when you need the report again.

With Bash, we can also:

- Pass a filename as an **argument**.
- Store values in **variables**.
- Choose and repeat commands with **conditions and loops**.

7/56

## Which tool fits the task?

Université de Rennes
Louis Ledoux

| Task | A useful starting point |
|---|---|
| Explore sales figures, filter and make a chart | Excel or another graphical analysis tool |
| Inspect a server's logs, processes and permissions | Remote shell and system commands |
| Apply the same sequence of commands to several files | A shell script with arguments and a loop |
| Query related business tables | SQL and a database |
| Explain the result to the team | A spreadsheet, dashboard or report |

Excel also offers Power Query and Office Scripts for repeatable work.

8/56

Université de Rennes
Louis Ledoux

# 1. Linux and the shell

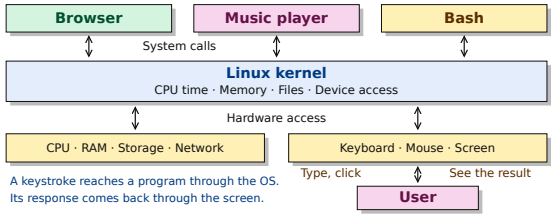
What happens between typing a command and the machine doing the work?

9/56

## Programs, kernel and hardware

Université de Rennes
Louis Ledoux

We use hardware to interact with programs.




The kernel **shares resources and controls access**.

10/56

## Linux

Université de Rennes
Louis Ledoux

- **1991:** Linus Torvalds announces his kernel project.
- Linux is a **Unix-like kernel**, distributed as free and open source software.
- A distribution combines the kernel with tools, libraries and applications.

Our working environment: a Linux system with the **Bash shell** and command-line tools.

11/56

## Unix, GNU and Linux

Université de Rennes
Louis Ledoux

| Name | Meaning |
|---|---|
| **Unix** | The operating-system family that inspired Linux |
| **GNU** | The project behind Bash and many command-line tools |
| **Linux** | The kernel: manages processes, memory and hardware |

**GNU = GNU's Not Unix.**
The name contains itself: a recursive acronym.

12/56

## Touch, click or type

Université de Rennes
Louis Ledoux

You already use interfaces to start programs.

| Your action | Interface | Example |
|---|---|---|
| **Touch** a phone screen | Graphical interface | Open an app from its launcher |
| **Click** with a mouse | Graphical interface | Open a folder in a file manager |
| **Type** a command | Text interface | Ask Bash to list a directory |

A smartphone can be a **terminal**: an endpoint for using a service.
Its touchscreen, like a mouse or keyboard, provides **input**.

13/56

## Our terminal for this course?

Université de Rennes
Louis Ledoux



DEC VT100 · IF-3-SYS, G. Salagnac, INSA Lyon

A **physical terminal** sends keystrokes to a computer and displays its replies.
The programs run on the connected computer.

Good news: yours fits in a window.

14/56

## Terminal, shell and command

Université de Rennes
Louis Ledoux

| Name | Role |
|---|---|
| **Terminal emulator** | A window for keyboard input and text output |
| **Bash, the shell** | Interprets the line and runs commands |
| **`ls`, a command** | Lists directory entries |

Type `ls` → Bash runs `ls` → the terminal displays its output.

**TTY** = *teletypewriter*: early terminals printed on paper.
Unix keeps the name, even for emulated terminals.

15/56

## A command and its arguments

Université de Rennes
Louis Ledoux

```
ls -l /home/alice
```

| Part | Role |
|---|---|
| `ls` | **Command**: list directory entries |
| `-l` | **Option**: change how `ls` lists them |
| `/home/alice` | **Operand**: the directory to list |

Both `-l` and `/home/alice` are **arguments** passed to `ls`.

16/56

## Short and long options

Université de Rennes
Louis Ledoux

| Form | Example | Meaning for GNU `ls` |
|---|---|---|
| Short: **one dash** | `ls -a` | Include hidden entries |
| Long: **two dashes** | `ls --all` | Same as `-a` |
| Combined | `ls -la` | `-l` and `-a` together |

Some options take a value: `head -n 5 notes.txt` prints five lines.

A **bare `--` ends options**.
`ls -- -notes.txt` treats `-notes.txt` as a filename.

Each command defines its options. Check `--help` or `man`.

17/56

## Getting help

Université de Rennes
Louis Ledoux

```
man ls
info coreutils
apropos directory
help cd
```

- **Tab** completes a name or displays possible completions.
- **Up** recalls a previous command. `history` lists history.
- `clear` clears the display. In `man`, **q** quits.

18/56

Université de Rennes
Louis Ledoux

# 2. Files and directories

Where is my file, and what can I do with it?

19/56

## One directory tree

Université de Rennes
Louis Ledoux

A **directory** is a folder: it contains names of files and other directories.

```
/
├── home/
│   └── alice/
│       └── notes.txt
└── etc/
```

The common root is /. The path `/home/alice/notes.txt` follows the tree.
Alice's **home directory** is `/home/alice`; the university may use a different path.

20/56

## Personal files and system data

Université de Rennes
Louis Ledoux

Linux separates personal files, configuration and application data.

| Directory | Typical contents |
|---|---|
| `/home/alice` | Alice's personal files |
| `/etc` | System configuration |
| `/var` | Changing data: logs, caches, application state |
| `/var/lib` | Persistent application state |
| `/var/log` | Log files |

`/var/lib` and `/var/log` are subdirectories of `/var`.

21/56

## Users, administrators and software

Université de Rennes
Louis Ledoux

| Directory | Typical role |
|---|---|
| `/home`, `/root` | User homes; the root account's home |
| `/usr` | Most distribution-provided tools and libraries |
| `/usr/local` | Software installed locally by the administrator |
| `/opt` | Add-on software packages |

/ is the filesystem root.
`root` is also the administrator account's name.

22/56

## Temporary files and system information

Université de Rennes
Louis Ledoux

| Directory | Purpose / lifetime |
|---|---|
| `/var` | Changing data, often persistent |
| `/tmp` | Temporary files; cleanup policy varies |
| `/run` | Runtime state, cleared at boot |
| `/dev` | Device interfaces |
| `/proc`, `/sys` | Views of processes, kernel and devices |

A path can expose live system information.
It need not refer to ordinary data stored on disk.

23/56

## Several filesystems, one tree

Université de Rennes
Louis Ledoux

```
/
├── home/       user files
├── mnt/        a temporary mount point
└── media/      removable media mount points
```

Mounting makes another filesystem visible at a directory.
The device node might be `/dev/sdb1`; its files appear at the mount point.

24/56

## The current directory

Université de Rennes
Louis Ledoux

| Notation | Meaning |
|---|---|
| `.` | Current working directory |
| `..` | Parent directory |
| `/` | Root directory |
| `~` | Your home directory, expanded by the shell |
| `$HOME` | Variable containing your home path |
| `$PWD` | Variable containing the current directory path |

25/56

## Absolute and relative paths

Université de Rennes
Louis Ledoux

Current directory: `/home/alice`

| Path | Interpretation |
|---|---|
| `/home/alice/notes.txt` | Absolute: start at `/` |
| `notes.txt` | Relative: start in the current directory |
| `./notes.txt` | The same relative path |
| `../bob/notes.txt` | Go to the parent, then into `bob` |

A path tells us **where to look**. Permissions determine whether access is allowed.

26/56

## Moving around

Université de Rennes
Louis Ledoux

```
pwd
cd /home/alice
cd ..
cd
```

- `pwd` prints the current working directory.
- `cd path` changes the current directory.
- `cd` or `cd ~` returns to your home.

27/56

## Listing directory entries

Université de Rennes
Louis Ledoux

```
ls
ls -l
ls -a
ls -la /home/alice
```

- `-l` adds details, including ownership and permissions.
- `-a` includes names beginning with `.`, such as `.bashrc`.
- `ls directory` lists that directory's contents.

28/56

## Creating and removing directories

Université de Rennes
Louis Ledoux

```
mkdir practice
rmdir practice
```

`rmdir` removes an **empty** directory.

`rm -r practice` also removes the directory's contents.
Shell deletion does **not** send files to a recycle bin.

29/56

## Creating, copying and moving files

Université de Rennes
Louis Ledoux

```
touch notes.txt
cp notes.txt notes-copy.txt
mv notes-copy.txt archive.txt
rm archive.txt
```

`touch` creates a missing file, or updates timestamps of an existing file.

30/56

## Reading text files

Université de Rennes
Louis Ledoux

| Command | Result |
|---|---|
| `cat notes.txt` | Print the whole file |
| `less notes.txt` | Browse interactively, **q** to quit |
| `more notes.txt` | Browse a page at a time |
| `head -n 5 notes.txt` | Print the first five lines |
| `tail -n 5 notes.txt` | Print the last five lines |
| `diff old.txt new.txt` | Show differences between two files |

31/56

## Why is less called less?

Université de Rennes
Louis Ledoux

Mark Nudelman was investigating errors in large logs.
He wanted to read the lines **before** an error.

His version of `more` could not move backward.
The new pager's name joked about a **backward more**.

```
less app.log
```

32/56

## Concatenating files

Université de Rennes
Louis Ledoux

```
cat first.txt second.txt
cat
```

- With filenames, `cat` prints their contents **in argument order**.
- Without filenames, `cat` reads **standard input**.
- At an empty terminal input line, **Ctrl+D** indicates end of input.

33/56

## Filename patterns

Université de Rennes
Louis Ledoux

| Pattern | Matches |
|---|---|
| `?` | Exactly one character |
| `*` | Zero or more characters |
| `[abc]` | One character from the set |

```
ls /etc/rc*
ls /etc/rc????
ls [abc]oo.c
```

The **shell** expands these patterns before running `ls`.

34/56

## Spaces separate arguments

Université de Rennes
Louis Ledoux

Our file is named **`meeting notes.txt`**, including the space.

**Without quotes**

```
cat meeting notes.txt
```

Two filename arguments:
`meeting` and `notes.txt`.
`cat` tries to open two files.

**With quotes**

```
cat "meeting notes.txt"
```

One filename argument:
`meeting notes.txt`.
`cat` opens the intended file.

The shell removes the quotes. They are **instructions for the shell**.

35/56

## Three ways to protect a space

Université de Rennes
Louis Ledoux

These commands all pass **the same filename** to `cat`:

```
cat 'meeting notes.txt'
cat "meeting notes.txt"
cat meeting\ notes.txt
```

- **Single quotes** preserve everything inside them literally.
- **Double quotes** preserve spaces, but still allow `$HOME` to expand.
- An unquoted **backslash** protects the next character, such as a space or `*`.

36/56

## What expands inside quotes?

Université de Rennes
Louis Ledoux

Suppose `$HOME` contains `/home/alice`.

| Command | Output |
|---|---|
| `echo '$HOME'` | `$HOME` |
| `echo "$HOME"` | `/home/alice` |
| `echo "$HOME/*.txt"` | `/home/alice/*.txt` |
| `echo "It's OK"` | `It's OK` |
| `echo 'Say "hi"'` | `Say "hi"` |

Double quotes expand the variable, but keep spaces and * literal.
The other kind of quote can appear as ordinary text inside quotes.

37/56

## Filename patterns: expanded or literal?

Université de Rennes
Louis Ledoux

The directory contains only **`a.txt`** and **`meeting notes.txt`**.

**Expand the pattern**

```
echo *.txt
```

```
a.txt meeting notes.txt
```

The shell finds matching filenames.

**Keep the pattern literal**

```
echo '*.txt'
```

```
*.txt
```

`echo "*.txt"` and `echo \*.txt` also print `*.txt`.

`echo` separates its arguments with spaces. Its output hides their boundaries.

38/56

## Seeing each argument with printf

Université de Rennes
Louis Ledoux

`printf '<%s>\n'` prints each following argument between < and >, on its own line.

```
printf '<%s>\n' *.txt
```

```
<a.txt>
<meeting notes.txt>
```

**Two arguments.** The space inside the matched filename stays intact.

```
printf '<%s>\n' '*.txt'
```

```
<*.txt>
```

**One argument.** The quoted pattern stays literal.

39/56

## Checkpoint: files and paths

Université de Rennes
Louis Ledoux

1. What does `cd ..` do?
2. Which `ls` option includes hidden entries?
3. What does `cat a.txt b.txt` print first?
4. Does `echo '*.txt'` expand the pattern?

1. Move to the parent directory.
2. `-a`
3. The contents of `a.txt`.
4. No. Single quotes preserve the literal *.

40/56

Université de Rennes
Louis Ledoux

# 3. Who may access this file?

Why can I read this file, but cannot change or delete it?

41/56

## Users, groups and ownership

Université de Rennes
Louis Ledoux

```
whoami
id
ls -ld .
ls -l
```

- `id` shows your user identity and groups.
- `ls -l` shows the file's owner, group and permission bits.
- `ls -ld` inspects a directory itself.

42/56

## Permission classes

Université de Rennes
Louis Ledoux

```
-rwxr-xr--
  u  g  o
```

| Class | Applies to |
|---|---|
| u | The file's owner |
| g | Members of the file's group |
| o | Other users |
| a | All three classes when using `chmod` |

43/56

## What r, w and x allow

Université de Rennes
Louis Ledoux

| Bit | Regular file | Directory |
|---|---|---|
| **r** | Read contents | List entry names |
| **w** | Modify contents | Add or remove entries* |
| **x** | Execute | Search / traverse |

*Changing directory entries also requires search permission. Other restrictions, such as the sticky bit, may apply.

44/56

## Changing permission bits

Université de Rennes
Louis Ledoux

```
chmod u+x script.sh
chmod go-w notes.txt
chmod u=rw,g=r,o= notes.txt
```

- `+` adds the selected permission bits.
- `-` removes them.
- `=` sets the selected class to the listed permissions.

45/56

## Numeric permissions

Université de Rennes
Louis Ledoux

Each class adds its bit values: **r = 4**, **w = 2**, **x = 1**.

| Owner | Group | Others |
|---|---|---|
| `rwx` = **7** | `rw-` = **6** | `r--` = **4** |

```
chmod 764 notes.txt
```

The three digits describe **owner, group, others**, in that order.

46/56

## Checkpoint: permissions

Université de Rennes
Louis Ledoux

1. Decode `640` for a regular file.
2. Add execute permission for the owner of `run.sh`.
3. Which directory bit allows traversal?

1. Owner: `rw-`
   Group: `r--`
   Others: `---`
2. `chmod u+x run.sh`
3. `x`

47/56

## “I can see the name, but cannot read it”

Université de Rennes
Louis Ledoux

```
# Illustrative server metadata
 drwxr----- alice students  shared/
 -rw-r----- alice students  shared/notes.txt
```

Bob belongs to `students`. He can list the names, but cannot open `notes.txt`. Why?

The directory’s group bits lack **x**: traversal.
If team access is intended, the owner can add `g+x`.

48/56

## CM1 summary: files and permissions

Université de Rennes
Louis Ledoux

1. Locate configuration, software, data and logs.
2. List and read files; quote names containing spaces.
3. Check owner, group and other permissions.

49/56

Université de Rennes
Louis Ledoux

## Optional reference

Examples to revisit during TD or personal practice

50/56

## Operating system, shell and GUI

Université de Rennes
Louis Ledoux

| Layer | What it gives us |
|---|---|
| **Operating system** | Processes, storage and access control |
| **Graphical interface** | Windows, menus and direct visual interaction |
| **Shell** | Commands we can combine and save in scripts |
| **Application** | A task-specific tool, such as Excel or a database |

Your laptop can run Windows while the company's application runs on Linux. You need to work with both.

51/56

## Summarise a log before opening it in Excel

Université de Rennes
Louis Ledoux

| Quantity | Meaning |
|---|---|
| **2,000,000 log lines** | Example input size |
| **1,048,576 rows** | The maximum in one Excel worksheet |
| **One row per error code** | The summary the team wants to discuss |

Process the raw log where it lives, save the counts, then use Excel to analyse or present the summary.

Microsoft: worksheet limits. Power Query and the Data Model offer other ways to handle larger datasets.

52/56

## Path components and word counts

Université de Rennes
Louis Ledoux

```
dirname /home/alice/notes.txt
basename /home/alice/notes.txt
wc -w notes.txt
```

- `dirname` prints `/home/alice`.
- `basename` prints `notes.txt`.
- `wc -w` counts **words**. `wc -l` counts newline characters.

53/56

## Hard links and symbolic links

Université de Rennes
Louis Ledoux

### Hard link

```
ln notes.txt other.txt
```

- Another name for the **same file**.
- Removing one name leaves the other usable.
- Same filesystem, normally no directory links.

### Symbolic link

```
ln -s notes.txt shortcut
```

- Stores a **path** to a target.
- May cross filesystems or target a directory.
- Can become dangling if the target path disappears.

54/56

## Builtins and external commands

Université de Rennes
Louis Ledoux

**Builtin**

Implemented inside the shell.

```
cd ..
```

Can change the current shell's state.

**External program**

Loaded from an executable file.

```
ls -l
```

The shell searches directories in PATH.

55/56

## Recognising a command

Université de Rennes
Louis Ledoux

```
type cd
type ls
type -a echo
help cd
```

`type` reports how Bash resolves a name.
`help` documents its builtins.

56/56